ModelIt!

Teacher's Guide

How Does Sound Travel?

The Science of Vibrations and Sound Waves

Quick Reference

Grade Level	Grade 1
NGSS Standard	1-PS4-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 1-PS4-1

Plan and conduct investigations to provide evidence that vibrating materials can make sound and that sound can make materials vibrate.

Three-Dimensional Learning

Science Practice	**Planning and Carrying Out Investigations** — Students investigate how changing vibration strength affects the loudness of sound.
Disciplinary Core Idea	**PS4.A Wave Properties** — Sound can make matter vibrate, and vibrating matter can make sound.
Crosscutting Concept	**Cause and Effect** — Students explore how stronger vibrations cause louder sounds.

CAST Testing Alignment

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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> **For Your Confidence**
>
> This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Sound is all around us every day. From birds singing to cars honking, all sounds are made by vibrations — things shaking back and forth very quickly.

What Makes Sound?

Every sound starts with a vibration. When you pluck a guitar string, it shakes back and forth very fast. When you hit a drum, the drum skin vibrates. These vibrations push the air around them, creating invisible waves that travel outward in all directions, like ripples when you drop a stone in water. When these air waves reach our ears, tiny parts inside our ears vibrate too, and our brain understands it as sound.

Loud and Soft Sounds

The strength of a vibration determines how loud a sound is. When you tap a drum gently, it vibrates a little and makes a soft sound. When you hit it hard, it vibrates a lot and makes a loud sound. Scientists call the size of a vibration its amplitude. Big amplitude means loud sound, and small amplitude means soft sound. Sound can travel through air, water, and even solid objects like walls and tables.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System
Students identify vibration strength as something we can control (external) and sound volume as something that changes in response (internal).

E — Establish Relationships
Students figure out that when vibration strength goes up, sound volume also goes up — a positive relationship.

V — Visualize & Model
Students build a simple model with two parts showing how vibration strength connects to sound volume.

E — Evaluate Outcomes
Students test their model by trying soft taps and hard taps and observing how the sound changes.

R — Revise & Extend
Students reflect on how their model matches what happens when they play real instruments or clap their hands.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Today we are going to explore something invisible but very powerful — SOUND! Can everyone clap their hands? You just made sound!"*

Watch For

Have students clap, snap, and stomp to make different sounds. Ask: How are we making these sounds?

Slide 2: What Will We Learn?

3 min

Say: *"We are going to find out how sound travels from one place to another. We will learn a new word: vibration!"*

Watch For

Introduce the word vibration. Have students place their hand on their throat and hum to feel vibrations.

Slide 3: Let's Be Scientists!

3 min

Say: *"Here is our big question: How does sound get from one place to another? What do you think?"*

Watch For

Give students 30 seconds to think, then share with a partner. Collect a few ideas from the class.

Slide 4: What Are the Parts?

5 min

Say: *"Our model has two parts: Vibration Strength is how hard something shakes. Sound Volume is how loud the sound is."*

Watch For

Hold up a drum or rubber band. Show a gentle pluck vs. a strong pluck. Ask: Which one can we control?

Slide 5: How Are They Connected?

5 min

Say: *"When vibration strength goes UP, what happens to the sound? Does it get louder or softer?"*

Watch For

Students use sentence frame: When vibration strength goes up, sound volume goes up. Draw the arrow together.

Slide 6: What Did We Find Out? 5 min

Say: *"We found out that bigger vibrations make louder sounds! Sound is just vibrations traveling through the air."*

Watch For

Review discoveries. Sprinkle rice on a drum and tap it so students can SEE the vibrations.

Slide 7: STEM Challenge 10 min

Say: *"Now it is YOUR turn to build an instrument! Can you make something that plays loud AND soft sounds?"*

Watch For

Distribute materials (rubber bands, cups, boxes, paper towel tubes). Students build and test instruments.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Vibration Strength
- **INTERNAL:** Sound Volume

Relationships

- **Vibration Strength to Sound Volume:** POSITIVE (+) — When you make stronger vibrations by tapping harder, the sound gets louder. Bigger shakes make bigger sounds.

Key Discoveries

- Bigger vibrations make louder sounds
- Sound travels through the air from the thing that vibrates to your ears
- You can feel vibrations if you touch something that is making sound

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Sound just happens by itself"	Sound never happens without a vibration. Something always has to shake or move back and forth to make sound. Even your voice comes from tiny parts in your throat vibrating!
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Sound only travels through air"	Sound can travel through solids and liquids too! Put your ear on a table and have a friend tap the other end — you can hear it through the wood. Whales sing to each other through water.